

Guide: Replacing an Orbit Prefix

Collatz proof project

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Why merging helps

Imagine the smallest positive integer with an unbounded Collatz orbit. If a smaller positive integer eventually joins that orbit, its orbit is also unbounded, contradicting the choice of the smallest one. This is a valid proof strategy only if we can guarantee that smaller predecessor. Lean now proves the implication; existence remains open.

What the replacement certificate checks

A finite odd/even pattern determines an affine endpoint formula. Two patterns with the same length and number of odd steps have the same multiplier. Increasing the correction by $3^j\delta$ reduces the starting seed by δ while keeping the endpoint fixed. The proof checks that the new integer actually follows every specified parity. An arbitrary number in a plausible correction interval is insufficient.

The replacement is useful when $0 < \delta < n$, so its seed is positive and smaller. It need not grow at every intermediate step.

A working infinite family

Any initial run of odd steps followed by two even steps can be replaced by a pattern starting from the integer immediately below the original seed. For example,

15	23	35	53	80	40	20
14	7	11	17	26	13	20

These are six shortcut steps with four odd steps on each row. This is the classical Garner stem family, now formalized in this project. It is not a new mathematical discovery or a universal coverage result.

Why this version cannot cover every seed

Lean proves that 27 never merges with a smaller positive seed at the same time with the same number of odd steps. This holds for every time, not just a search cutoff. The proof checks the initial 70 steps and then uses the repeating cycle: 27 has accumulated too many odd steps for any smaller seed to catch up under these two restrictions. Since 27 reaches one, this does not disprove the proposed strategy for a hypothetical least unbounded seed. It does mean that an unrestricted claim of equal-time, equal-count replacement for every seed would be false.

What the experiment establishes

At depth 18, exact enumeration finds replacements for 193,309 of all 262,144 binary cylinders. Among the 7,495 patterns whose every prefix coefficient is at least one, it covers 1,391. The uncovered patterns are not divergent orbits. The counts do not prove asymptotic or universal coverage.

Reproduce and continue

From `lean/`, run `lake build Collatz.WordSurgery`. From the root, run `python3 analysis/word_surgery.py` and `python3 -m unittest discover -s analysis -p test_word_surgery.py`. Different word lengths or odd counts may help, but neither nondivergence nor exclusion of all nontrivial cycles has been proved here.